

Energy–Parameterized Credit-assignment (ePC): Experimental Report – R8 Broad Outcome & R9 Stage A Representation Battery

ZeroBot Research Agent
for Benjamin Goertzel

2026-07-22

Abstract

This report documents two experiment batches on Energy–Parameterized Credit-assignment (ePC), a training method that enforces an energy–monotonicity property during knowledge distillation. Experiment R8 evaluates raw predictive performance across 5 seeds and 5 arms at both matched-update and matched-wall-clock budgets. Experiment R9 Stage A probes the learned representations along four axes: calibration validity (A0), representational collapse (A1), factor probe accuracy (A2), and causal selectivity (A3). The headline finding is that ePC’s energy–monotonicity property is robustly validated (10/10 EPC records), but ePC does not outperform plain baselines on validation loss, and its learned representations exhibit significantly more collapse and faster loss of task-relevant factor information.

Contents

1	Background and Motivation	2
2	Experiment R8: Broad Outcome	2
2.1	Setup	2
2.2	R8 Results: Per-Seed Records	2
2.3	R8 Results: Aggregate Statistics	3
2.4	R8 Key Findings	3
2.5	Credit Assignment Profile (EPC arms)	4
3	Experiment R9 Stage A: Representation Battery	4
3.1	Setup	4
3.2	A0: Calibration Validation	4
3.3	A1: Representational Collapse	5
3.4	A2: Factor Probe Accuracy	6
3.5	A3: Causal Selectivity	6
4	Synthesis and Interpretation	6
4.1	Convergent Evidence	6
4.2	Mechanistic Read	7
4.3	What Is Validated vs What Is Not	7
4.4	Recommendation	7
5	Provenance	8

1 Background and Motivation

ePC (Energy-Parameterized Credit-assignment) is a training objective that introduces an energy-based constraint into the knowledge distillation (KD) process. The core idea is that credit assignment to deeper network layers should respect an energy-monotonicity property: the “activity energy” of each layer should be non-increasing along the inference path, providing a structured alternative to ordinary backpropagated KD gradients.

Two variants are tested:

- **epc_original**: standard ePC with default inference depth.
- **epc_deep**: ePC with increased inference depth ($T=12$).

These are compared against three baselines:

- **bp_ce**: ordinary backpropagation with cross-entropy loss.
- **bp_kd**: ordinary backpropagation with KD loss, matched at 200 updates.
- **bp_kd_wallclock**: ordinary KD allowed to run for the same wall-clock budget as epc_original (~ 445 s), completing $\sim 6.7\times$ more update steps.

The teacher model is a pretrained GPT-2 small (val_loss 4.124, val_perplexity 61.8). The student is a 6-layer GPT-2 trained on a synthetic subject-verb-object grammar corpus.

2 Experiment R8: Broad Outcome

2.1 Setup

- **Pod**: RunPod A100 PCIe 80GB, \$1.39/hr
- **Source commit**: ecf2f79
- **Seeds**: 1729, 3253, 6421, 8191, 10103 (5 seeds)
- **Arms**: bp_ce, bp_kd, epc_original, epc_deep, bp_kd_wallclock (5 arms)
- **Total records**: 25 (5 seeds $\times$ 5 arms)
- **Matched-update arms**: 200 updates each (bp_ce, bp_kd, epc_original, epc_deep)
- **Matched-wall-clock arm**: bp_kd_wallclock runs for ~ 445 s (same elapsed budget as epc_original), completing ~ 1334 updates on average.

2.2 R8 Results: Per-Seed Records

Arm	Seed	val_loss	val_ppl	kd_gap	updates	elapsed (s)
bp_ce	1729	6.265	525.7	1158.8	200	130
bp_ce	3253	6.194	489.8	1164.6	200	64
bp_ce	6421	6.222	503.9	1148.5	200	65
bp_ce	8191	6.270	528.3	1163.6	200	65
bp_ce	10103	6.353	574.0	1169.0	200	64
bp_kd	1729	7.053	1156.5	828.1	200	104
bp_kd	3253	7.053	1155.8	830.3	200	66
bp_kd	6421	6.950	1043.3	807.6	200	67
bp_kd	8191	7.037	1137.5	818.6	200	67
bp_kd	10103	7.103	1215.2	834.3	200	66
epc_original	1729	6.747	851.4	981.0	200	591
epc_original	3253	6.656	777.7	987.9	200	406
epc_original	6421	6.695	808.1	962.4	200	414
epc_original	8191	6.777	877.5	992.0	200	409
epc_original	10103	7.028	1128.3	966.5	200	403
epc_deep	1729	6.659	780.1	941.3	200	2032
epc_deep	3253	6.615	746.5	953.6	200	1673

Arm	Seed	val_loss	val_ppl	kd_gap	updates	elapsed (s)
epc_deep	6421	6.713	823.0	943.1	200	1670
epc_deep	8191	6.869	961.8	970.9	200	1660
epc_deep	10103	6.913	1004.9	994.1	200	1655
bp_kd_wallclock	1729	5.587	267.0	418.9	1783	591
bp_kd_wallclock	3253	5.824	338.4	482.2	1220	406
bp_kd_wallclock	6421	5.711	302.3	465.6	1229	414
bp_kd_wallclock	8191	5.881	358.3	491.1	1225	409
bp_kd_wallclock	10103	5.918	371.7	491.0	1213	403

Table 1: R8 per-seed per-arm results. Teacher: val_loss=4.124, val_ppl=61.8.

2.3 R8 Results: Aggregate Statistics

Arm	val_loss (mean $\pm$ sd)	val_ppl	kd_gap	updates	elapsed (s)
bp_ce	6.261 $\pm$ 0.060	524.3	1160.9	200	78
bp_kd	7.039 $\pm$ 0.056	1141.6	823.8	200	74
epc_original	6.781 $\pm$ 0.146	888.6	978.0	200	445
epc_deep	6.754 $\pm$ 0.131	863.2	960.6	200	1738
bp_kd_wallclock	5.785 $\pm$ 0.135	327.5	469.8	1334	445
<i>Teacher</i>	<i>4.124</i>	<i>61.8</i>	—	—	—

Table 2: R8 aggregate statistics (5 seeds per arm).

2.4 R8 Key Findings

1. **Energy-monotonicity holds robustly.** All 10 EPC records (5 seeds $\times$ {epc_original, epc_deep}) have activity_energy_monotone: true. This is the primary positive structural finding.
2. **At matched update count (200), bp_ce beats both EPC arms.** bp_ce val_loss 6.261 vs epc_original 6.781 and epc_deep 6.754. EPC does not improve over plain cross-entropy at equal updates.
3. **At matched wall-clock, ordinary KD dominates.** bp_kd_wallclock completes $\sim 6.7\times$ more updates (1334 vs 200) and achieves val_loss 5.785 / val_ppl 327.5, far better than all 200-update arms.
4. **epc_deep is marginally better than epc_original** on val_loss (6.754 vs 6.781), but the difference is within seed variation (± 0.13). Increasing inference depth from $T=6$ to $T=12$ does not yield a clear performance gain.
5. **bp_kd at 200 updates is the worst arm** (val_loss 7.039). Plain KD at low update counts produces worse students than plain CE, consistent with the KD signal being noisy when undertrained.
6. **EPC is substantially slower.** epc_original averages 445s per run; epc_deep averages 1738s ($3.9\times$ slower than epc_original, $22\times$ slower than bp_ce). The energy-monotonicity enforcement requires iterative inference at each step.

2.5 Credit Assignment Profile (EPC arms)

The credit assignment diagnostic records the ePC gradient norm and ordinary KD gradient norm per transformer block. In both EPC arms, the ePC gradient norm is effectively zero for blocks 0–4 and only non-zero for the final block (h.5):

Block	ePC norm	ordinary KD norm	norm ratio	finite
h.0	0.0000	28715.22	0.000000	True
h.1	0.0000	22168.94	0.000000	True
h.2	0.0000	18854.66	0.000000	True
h.3	0.0000	16373.57	0.000000	True
h.4	0.0000	14862.22	0.000000	True
h.5	0.4913	15015.60	0.000033	True

Table 3: Credit assignment for seed 3253, epc_original. epc_deep shows an identical pattern (ePC norm 0.4916 at h.5 only).

This means the ePC signal is concentrated entirely in the last transformer block, with the ratio ePC/KD $\approx 3.3 \times 10^{-5}$. The energy-monotonicity constraint is structurally enforced, but the credit it assigns is negligible relative to ordinary KD gradients.

3 Experiment R9 Stage A: Representation Battery

3.1 Setup

R9 Stage A was run on the same pod immediately after R8 training completed. It applies four probe families to the 20 checkpoints from the 4 matched-update arms (bp_ce, bp_kd, epc_original, epc_deep $\times$ 5 seeds):

- **A0 – Calibration:** CKA symmetry, self-CKA, known-rank recovery, permutation null accuracy.
- **A1 – Collapse:** entropy effective rank, participation ratio, top-1 variance fraction, dead neuron fraction. Measured at embedding, 6 transformer blocks, and final layer.
- **A2 – Factor Probes:** linear classifier accuracy (test and held-out) for four syntactic factors: subj_num, obj_num, tense, negation.
- **A3 – Causal Selectivity:** intervention-based selectivity score for the same four factors.

3.2 A0: Calibration Validation

Metric	Value
CKA symmetry error	6.55×10^{-15}
CKA (X,Y)	0.958
CKA (self)	1.000
Known rank (input)	5
Known rank (recovered)	5
Permutation null train acc	0.531
Permutation null test acc	0.448

Table 4: A0 calibration (seed 3253, epc_original). All probes pass.

CKA symmetry error is at machine-precision zero, self-CKA is exactly 1.0, and known-rank recovery is exact. The probe infrastructure is trustworthy.

3.3 A1: Representational Collapse

This is the most diagnostically informative probe. The effective rank measures the number of significant dimensions in the representation; the participation ratio is a complementary measure of dimensionality; top-1 variance fraction measures how much of the total variance is captured by the single leading principal component.

Layer	bp_ce	bp_kd	epc_orig	epc_deep
<i>Entropy Effective Rank</i>				
embedding	374.65	365.65	342.79	340.98
block_0	51.78	7.21	8.18	7.84
block_1	30.60	5.71	3.70	3.62
block_2	21.18	5.15	2.80	2.64
block_3	16.63	4.88	2.58	2.40
block_4	14.06	4.74	2.49	2.32
block_5	12.33	4.66	3.38	3.18
final	13.90	4.98	3.87	3.78
<i>Participation Ratio</i>				
embedding	186.77	169.86	129.88	126.88
block_0	13.70	3.43	4.15	3.99
block_1	9.83	3.21	2.38	2.32
block_2	7.77	3.12	1.91	1.80
block_3	6.65	3.05	1.80	1.67
block_4	5.95	3.00	1.77	1.63
block_5	5.42	2.98	2.28	2.15
final	6.14	3.13	2.62	2.59
<i>Top-1 Variance Fraction</i>				
embedding	0.0354	0.0418	0.0595	0.0610
block_0	0.1993	0.4480	0.4121	0.4230
block_1	0.2515	0.4710	0.6106	0.6211
block_2	0.2964	0.4773	0.7038	0.7294
block_3	0.3285	0.4876	0.7296	0.7626
block_4	0.3528	0.4981	0.7373	0.7734
block_5	0.3746	0.5033	0.6250	0.6475
final	0.3406	0.4899	0.5663	0.5685
<i>Dead Neuron Fraction</i>				
all layers	0.0000	0.0000	0.0000	0.0000
final	0.0013	0.0013	0.0013	0.0013

Table 5: A1 representational collapse metrics (mean over 5 seeds).

Interpretation: EPC arms show dramatically more representational collapse. At the final layer:

- Effective rank: bp_ce 13.9, bp_kd 5.0, epc_original 3.9, epc_deep 3.8. EPC representations live in a ~ 3.8 -dimensional subspace.
- Top-1 variance: epc_original 0.566, epc_deep 0.569, vs bp_ce 0.341. A single component explains $\sim 57\%$ of EPC variance vs $\sim 34\%$ for CE.
- The collapse is not neuron death (dead fraction is identical at 0.13% across all arms). It is a low-dimensional *representational* bottleneck.
- bp_kd also shows significant compression (eff. rank 5.0), but not as severe as EPC.

- `epc_deep` is slightly more collapsed than `epc_original`, suggesting that increasing inference depth intensifies the compression rather than mitigating it.

3.4 A2: Factor Probe Accuracy

Linear classifiers are trained on each layer’s hidden states to predict four syntactic factors. Test accuracy on held-out sentences measures whether the factor is linearly decodable; held-out accuracy on unseen factor combinations measures compositional generalization.

Arm	subj_num		obj_num		tense		negation	
	test	heldout	test	heldout	test	heldout	test	heldout
<i>Final layer</i>								
<code>bp_ce</code>	0.957	0.903	0.696	0.257	0.913	0.856	1.000	1.000
<code>bp_kd</code>	0.838	0.732	0.688	0.120	0.824	0.763	0.987	0.978
<code>epc_original</code>	0.750	0.658	0.671	0.139	0.771	0.769	0.983	0.984
<code>epc_deep</code>	0.688	0.610	0.678	0.072	0.752	0.752	0.977	0.976
<i>Block 3</i>								
<code>bp_ce</code>	0.998	0.995	0.769	0.457	0.991	0.973	1.000	1.000
<code>bp_kd</code>	0.988	0.961	0.696	0.238	0.935	0.885	1.000	0.999
<code>epc_original</code>	0.500	0.503	0.511	0.482	0.537	0.554	0.876	0.860
<code>epc_deep</code>	0.506	0.516	0.565	0.340	0.637	0.640	0.880	0.874

Table 6: A2 factor probe accuracy (mean over 5 seeds). 0.5 = chance.

Interpretation: At block 3, EPC arms are at chance (0.50) for `subj_num`, while `bp_ce` retains 0.998. By the final layer, EPC partially recovers (0.69–0.75) but still trails `bp_ce` (0.957). The same pattern holds for `tense` and `obj_num`. Only `negation` is equally well-retained across all arms. `epc_deep` is slightly worse than `epc_original`, again suggesting depth intensifies information loss rather than helping.

3.5 A3: Causal Selectivity

Causal selectivity measures the degree to which intervening on a syntactic factor causally changes the representation, as measured by CKA discrepancy. Higher means more selective encoding of the factor.

Interpretation: All arms start at similar selectivity at the embedding layer (0.64–0.73 across factors), confirming the input representation is identical. The divergence begins at block 1 and widens through the depth of the network:

- **subj_num at block 3:** `bp_ce` 0.719, `bp_kd` 0.670, `epc_original` 0.399, `epc_deep` 0.375. EPC loses more than half the selectivity signal by mid-network.
- **tense at block 3:** `bp_ce` 0.749, `epc_original` 0.557. Same trend.
- **negation at block 3:** `bp_ce` 0.909, `epc_original` 0.581. Even the most robust factor degrades sharply.
- **epc_deep is consistently $\leq$ epc_original** on selectivity, confirming that deeper inference intensifies information loss.

4 Synthesis and Interpretation

4.1 Convergent Evidence

Three independent lines of evidence converge on the same conclusion:

1. **R8 Performance:** EPC does not beat baselines on raw validation loss at matched updates (6.75–6.78 vs 6.26 for CE) and is dominated at matched wall-clock (327 vs 863–889 perplexity).
2. **R9-A1 Collapse:** EPC representations are severely compressed (final effective rank 3.8–3.9 vs 13.9 for CE). A single component captures 57% of variance.
3. **R9-A2/A3 Factors:** EPC loses syntactic factor information faster than baselines. At block 3, factor probes are at chance for subj_num (0.50) and causal selectivity drops to 0.38–0.40 vs 0.72 for CE.

4.2 Mechanistic Read

The credit assignment diagnostic (Table 3) shows that the ePC gradient signal is effectively zero for blocks 0–4 and only non-zero at the final block (h.5), with a norm ratio of $\sim 3.3 \times 10^{-5}$ relative to ordinary KD. This means the energy–monotonicity constraint is structurally enforced but its credit signal is negligible for the lower layers.

The most likely mechanism is that ePC acts as an excessively strong attractor/regularizer: by enforcing energy monotonicity through iterative inference, it drives the network toward a low-energy fixed point that compresses the representation space. The constraint is satisfied (hence `activity_energy_monotone: true`), but at the cost of discarding task-relevant degrees of freedom.

The fact that `epc_deep` ($T=12$) is consistently equal to or slightly worse than `epc_original` on every metric — performance, collapse, factor retention, and selectivity — confirms that increasing inference depth does not reveal a hidden advantage. It intensifies the compression.

4.3 What Is Validated vs What Is Not

Validated:

- The energy–monotonicity property is real and robustly enforceable (10/10 EPC records across 5 seeds).
- The probe infrastructure is trustworthy (A0 calibration passes at machine precision).

Not supported:

- EPC does not improve predictive performance over plain baselines.
- EPC does not produce cleaner or more disentangled representations; it produces more collapsed ones.
- EPC does not preserve causal factor information better; it loses it faster.
- Increasing inference depth does not help.

Caveats:

- These results are for a 6-layer GPT-2 student on a synthetic grammar corpus. Whether the pattern holds at larger scale or on natural language is not tested.
- Stage A probes measure linear decodability and causal selectivity, not all possible representational properties.
- The ePC credit signal being concentrated in the final block suggests the current implementation may not be distributing credit as intended; a modified formulation could behave differently.

4.4 Recommendation

The current ePC formulation does not warrant advancement toward an adaptive residual layer. The structural property is valid but the practical effect is counterproductive: it compresses representations and discards factor information without a compensating performance or generalization benefit.

If the energy–monotonicity idea is to be pursued further, the next step should be diagnosing why the ePC credit signal is negligible for all but the final block, and exploring formulations that distribute credit more broadly before investing in additional scale or depth.

5 Provenance

- **R8 experiment:** `projects/relaleap/experiments/20260720T193211Z-epc-r8-broad-outcome/`
- **R8 artifacts:** `.../artifacts/r8/results/r8/train/training_summary.json` (25 records, schema `relaleap.epc_r8_train.v1`)
- **R9 Stage A experiment:** `projects/relaleap/experiments/20260721T222400Z-r9-stage-a/`
- **R9 artifacts:** `.../artifacts/seed*.json` (20 per-seed files) + `summary.json`
- **Source commit:** `ecf2f79`
- **Pod:** RunPod `0i26h131s399x4` (relaleap-r8-r9), A100-SXM4-80GB, \$1.49/hr
- **Seeds:** 1729, 3253, 6421, 8191, 10103
- **Teacher:** GPT-2 small (val_loss 4.124, val_ppl 61.8)
- **Student:** 6-layer GPT-2
- **Corpus:** synthetic subject-verb-object grammar
- **Report date:** 2026-07-22

Layer	bp_ce	bp_kd	epc_orig	epc_deep
<i>subj_num selectivity</i>				
embedding	0.645	0.648	0.645	0.645
block_0	0.793	0.763	0.757	0.765
block_1	0.783	0.732	0.558	0.573
block_2	0.757	0.708	0.424	0.407
block_3	0.719	0.670	0.399	0.375
block_4	0.672	0.633	0.408	0.400
block_5	0.620	0.606	0.507	0.401
final	0.607	0.514	0.438	0.379
<i>obj_num selectivity</i>				
embedding	0.670	0.666	0.663	0.663
block_0	0.578	0.507	0.531	0.530
block_1	0.557	0.491	0.348	0.352
block_2	0.518	0.468	0.283	0.265
block_3	0.496	0.453	0.343	0.350
block_4	0.461	0.404	0.358	0.344
block_5	0.425	0.357	0.301	0.254
final	0.425	0.302	0.232	0.275
<i>tense selectivity</i>				
embedding	0.719	0.726	0.730	0.730
block_0	0.834	0.809	0.784	0.780
block_1	0.813	0.766	0.618	0.610
block_2	0.789	0.731	0.552	0.510
block_3	0.749	0.704	0.557	0.509
block_4	0.707	0.657	0.490	0.448
block_5	0.674	0.627	0.566	0.511
final	0.661	0.592	0.546	0.481
<i>negation selectivity</i>				
embedding	0.685	0.680	0.676	0.676
block_0	0.891	0.868	0.863	0.874
block_1	0.899	0.863	0.764	0.796
block_2	0.908	0.848	0.662	0.712
block_3	0.909	0.815	0.581	0.592
block_4	0.904	0.767	0.569	0.566
block_5	0.887	0.713	0.749	0.754
final	0.870	0.720	0.719	0.718

Table 7: A3 causal selectivity scores (mean over 5 seeds).